



Woldia Institute of Technology (WiT)

School of Computing

Department of Computer Science

Exam Type: Model Exit Exam

Target Group: Computer Science GC students

Time Allowed: 2:30 Hr.

Full Name _____ **ID No.** _____ **Section** _____

General Instructions

- Switch off your mobile phone
- Check the exam paper contains 6 pages including this cover page
- Write your answers neatly and precisely
- Choose the best answer from the given alternatives and write your answer on the provided spaces only (Don't mark any answer on the question paper, if do that your answer will not be marked)
- Do not start until you are told to do so

“Good Work”

I. (ARTIFICIAL INTELLIGENT MODEL EXIT EXAM)

1. Which one of the following is incorrect about the goal of AI?
 - A) Develop intelligent agents.
 - B) construction of rational agents
 - C) provide capability of observing, learning, remembering & reasoning for machine
 - D) None**
2. Which one of the following is true statement about the type of AI?
 - A. Weak AI aims to create an agent that has complete humans intelligence
 - B. Weak AI aims to invent a machine called a mind in its fullest sense
 - C. Strong AI aims to create a machines which are conscious as human brains**
 - D. None
3. In Artificial Intelligence(AI), the simple reflex agent selects action on the basis of
 - A. Current percept, completely ignoring rest of the percept history.**
 - B. Rest of the percept history, completely ignoring current percept.
 - C. Both current percept and complete percept history
 - D. Both current percept and just previous history
4. The Task Environment of an agent consists of _____
 - A. Sensors
 - B. Actuators
 - C. Performance Measures
 - D. All of the mentioned**
5. Categorize Crossword puzzle in Fully Observable / Partially Observable.
 - A. Fully Observable**
 - B. partially Observable
 - C. All of the mentioned
 - D. None of the mentioned
6. The performance of an agent can be improved by _____
 - A. Learning**
 - B. Observing
 - C. Perceiving
 - D. None of the mentioned
7. Which is used to provide the feedback to the learning element?
 - A. Critic**
 - B. Actuators
 - C. Sensor
 - D. None of the mentioned
8. ____Environment is random in nature& can't be determined completely by an agent.
 - A. Deterministic
 - B. Stochastic**
 - C. Static
 - D. Dynamic
9. _____ is an agent that works based on the percept and the stored internal state.
 - A. Goal-based agents
 - B. Learning agents

C. Model-based reflex agents

D. Utility-based agents

10. A knowledge-based agent must be able to do:

- A. represent states, actions
- B. incorporate new percepts
- C. update the internal representation of the world

D. All

11. Knowledge engineering is the process of

- A. Extracting the knowledge from the human expert.
- B. Choose knowledge representation formalism
- C. Choose reasoning and problem solving strategy.

D. All

12. Which modifies the performance element so that it makes better decision?

- A. Performance element
- B. Changing element

C. Learning element

D. None of the mentioned

II. (NETWORKING MODEL EXIT EXAM)

13. Which OSI layer defines the functions of logical network addressing and routing?

- A. physical
- B. **network**
- C. data link
- D. transport

14. Dialogue and conversation between the sender and the receiver device is established by

- A. Physical
- B. **Session**
- C. data link
- D. presentation

15. Which one of the following is the main function of layer 4 of TCP/IP model?

- A. Provides interface between the applications and clients
- B. It performs the function of data compression, encryption, decryption and decompression
- C. It performs the function of data streaming
- D. **All of the above**

16. Which of the following devices translates hostnames into IP addresses?

- A. **DNS Server**
- B. Hub
- C. DHCP Server
- D. Firewall

17. Ethernet is _____

- A. one of the physical media
- B. it is WAN technology
- C. **it is LAN technology**
- D. client server network

18. Delivery and storage of E-Mail message to a server is achieved using _____

- A. post office protocol (POP)
- B. Internet mail access protocol (IMAP)
- C. **Simple mail transfer protocol (SMTP)**
- D. Hypertext transfer protocol (HTTP)

19. Changing a stream of data bits into a predefined code is known as _____

A. Decoding

B. Encoding

C. Signaling

D. Data transferring

20. Which one of the following is true statement about UDP?

A) UDP is connectionless protocol used for long messages

B) UDP has the response to the sender side

C) E-mail is an application used by UDP

D) DNS is an application used by UDP.

21. If we want to remove the message from mail box , which protocol is available to remove it

A. SMTP

B. IAMP

C. POP3

D. SNMP

22. The data rate which can be transmitted across a network can be measured by

A. Bit

C. Decimal

B. Bit per second

D. all

23. One of the following is incorrect about LAN

A. LAN is designed to operate for small physical area

B. It usually publicly owned network

C. In LAN sharing of resources becomes easy

D. Networks in LAN becomes easy to fix errors

24. A unique identifier that manufacturers assign to a network card or device is.

A. IP address

C. IPV4

B. MAC address

D. IPV6

III. (Algorithm Analysis model of exit exam)

25. Which one of the following is an incorrect statement about algorithms?

A. algorithm is composed of a finite set of steps

B. each steps of algorithm may require one or more operations

C. each operation has to be effective but not necessarily definite

D. All

26. Which one is incorrect statement about primitive and non-primitive data structures?

A. Primitive data structures encapsulate non primitive data structure to form more complex data structure

B. Arrays can be considered as a primitive data structure

C. Non primitive data structures are simpler than primitives

D. All

27. Which one of the following is correct about running time analysis?

- A. Determine how running time increases as the computer resources increases
 - B. Determine how running time increases as the size of the problem increases**
 - C. Determine how running time decreases as the computer resources increases
 - D. None
28. Which one of the following is correct about comparing algorithms?
- A. Comparing based on execution times is good for all cases
 - B. Comparing based on number of statements helps to compare universally
 - C. Express running time as a function of the input size n is ideal**
 - D. All
29. Which one of the following is correct about greedy algorithms?
- A. In each phase the currently best decision is made
 - B. optimal solution is a feasible solution where the objective function reaches its maximum or minimum
 - C. A globally optimal solution is one where there are no other feasible solutions with better objective function values.
 - D. All**
30. Which one of the following is incorrect about principle of optimality?
- A. an optimal sequence of decisions, some subsequence may not be optimal**
 - B. an optimal sequence of decisions, each subsequence must also be optimal
 - C. we need to show principle of optimality applies to the problem before using dynamic programming
 - D. ALL
31. Which one of the following is correct about asymptotic analysis?
- A. shorthand way to represent the time complexity
 - B. we can give time complexity as “fastest possible”, “slowest possible”
 - C. Rough measure that characterizes how fast each function grows.
 - D. All**
32. Which one of the following is a correct asymptotic notation?
- A. Big Ω is asymptotic upper bound
 - B. Big O is asymptotic lower bound
 - C. Big Θ is asymptotic tight bound**
 - D. All
33. Which one of the following is correct about binary trees (BT)?

A. A node may have at most have 2 child nodes

B. Complete BT all nodes are either a leaf or has degree 2.

C. Full BT all leave nodes are on the same level and all internal nodes have degree 2

D. All

34. Which one of the following is correct about heap data structure?

A. A heap is a nearly complete binary tree

B. all levels are always full

C. the last level may not be full, as heap is filled in order right to left

D. None

35. Considering the following array choose the correct max-heap representation.

4 1 3 2 16 9 10 14 8 7

A. 16, 14, 10, 9, 8, 7, 4, 3, 2, 1

C. 16, 14, 10, 8, 7, 9, 3, 2, 4, 1

B. 16, 10, 14, 7, 9, 8, 3, 2, 4, 1

D. 1, 2, 3, 4, 7, 8, 9, 10, 14, 16

36. Which one of the following is a correct time function $T(n)$ for binary search?

A. $T(n) = T(n-1) + O(n)$

C. $T(n) = T(n/2) + O(1)$

B. $T(n) = T(n/2) + T(n/2) + O(n)$

D. None

IV. (OOP MODEL EXIT EXAM)

37. Which one of the following is false about abstract classes? _____

A. Abstract classes are declared using the keyword abstract.

B. An abstract class can be a superclass.

C. An abstract class can also be a final class.

D. An object of an abstract class cannot be created.

38. Which one of the following is true about interfaces? _____

A. An interface's methods are by default public and final.

B. An interfaces variable is by default private, static and final.

C. An interface's methods are by default public and abstract.

D. An interface cannot be part of the inheritance hierarchy.

39. When a subclass object is created _____.

A. The very first constructor called is the subclass' constructor.

B. Any one of the constructors across the hierarchy can be called in a random order.

C. The last constructor called is the constructor of the superclass' constructor.

D. The very first constructor called is the top most superclass' constructor in the hierarchy.

40. When a method is overridden _____.

A. Both methods have exactly same signature.

B. Java uses it to implement compile time polymorphism.

C. The super keyword is used to call the overriding (subclass') method.

D. The overridden method must be final always so that the subclass can override it.

41. A reference variable can refer to _____.
A. An object of the superclass
B. An object of its class type only.
C. An object of any class.
D. An object of subclass's or its class type
42. _____ Refers to the process of assigning the address of the subclass' object to a superclass' reference variable.
A. Down casting
B. Up casting
C. Overloading
D. Overriding
43. When a class must not be inherited, it must be declared using the _____ keyword.
A. this
B. super
C. final
D. abstract
44. To call the superclass' constructor being inside a subclass' constructor the _____ keyword is used.
A. this
B. super
C. final
D. abstract
45. The keyword used to indicate that an interface is a superclass of another interface is _____.
A. implements
B. extends
C. implementing
D. interface

46. Which one of the following is false about protected members?
- A. Protected members can be accessed in the package they reside in.
 - B. Protected members can be accessed in any subclass of their container class.
 - C. A subclass can override protected members.
 - D. Protected members can be accessed in non-subclasses found in other packages.**

V. Software Engineering

Woldia Referral Hospital wants to use surgery management system. This system will help the Doctor by identifying the surgery activities based on the pre-stored information and give appropriate message immediately when the doctor misses any activity related to surgery.

According to the above case study, Answer question number 47 and 48.

47. The software developer team must be given more emphasis on which types of software quality requirements?
- A. Flexibility
 - B. Reliability**
 - C. Modifiability
 - D. Maintainability
48. What types of software process model is using more fit with the system requirement?
- A. Agile model
 - B. Spiral model**
 - C. Rational Unified Process Model
 - D. Waterfall model

Ethiopian Airline Company is decided to use automated human resource management system. Assume you are working in Alsam Software Development Company (ASDC). ASDC agrees with the Ethiopian Airline Company to develop this system. The Ethiopian Airline Company administrator says that “the implementation language must be Csharp” but the software developer team want to implement the system by java programming language.

Answer question number 49 and 50 based on the above case study

49. What types of requirement asked by Ethiopian Airline Company Administrator?
- A. Quality requirement
 - C. Pseudo requirement**
 - B. Functional requirement
 - D. Analysis requirement
50. Assume you are project manager, what is your decision?
- A. Accept the software development team idea and the system will be implemented by java.
 - B. Discuss with both Ethiopian airline administrator and software developer team to select common software programming language. If does not agree, the system development stop at early stage.**
 - C. Decide to stop system development automatically.
 - D. According to software engineering rule, the software developer team idea has more acceptance than clients. Therefore, the system will be implemented by java.
51. Which one of the following is false about UML object model can be mapped to relational database during software design phase?
- A. Each class is mapped to its own table
 - B. Each class attribute is mapped to a column in the table
 - C. Each class method is mapped to a relationship in the table**

- D.** An instance of a class represents a row in the table
52. Assume below two sentence are described non-functional requirements of Woldia university Student Information Management System (SIMS). What types of problem have been between those two non-functional requirement statements?
“The system must work 24 hours per day, 7 days per week and 365 days per year”
“The system can maintain within 3 hours ”
- A. Consistency** **C. Traceability**
B. Completeness **D. Realism**
53. Which one of the following statement is not described requirement elicitation is difficult?
- A.** The customers specify unnecessary technical detail that may confuse, rather than clarify, overall system objectives.
B. The customers are not completely sure of what is needed.
C. The requirements change over time
D. All customers have full information about their task and they can describe briefly.
54. Which one of the following true about design-level classes?
- A.** Design-level classes help to understand and document the problem space.
B. Design-level classes combine domain motivations with technical aspects which work in the solution space.
C. Design-level classes map directly to the corresponding construct of the implementation language.
D. Design-level classes consists attributes, methods and their relationships only.
55. Which one of the following false about model from software development perspective?
- A.** A model is a simplification of reality.
B. A model is an abstraction of something for the purpose of understanding, be it the problem or a solution.
C. Model is an abstraction of something for the purpose of understanding, which is applied during software design phase only.
D. A model is used to understand why a software system is needed, what it should do, and how it should do it.
56. “The system consists of more than one subsystem. When one subsystem modifies that highly affect on the other subsystems”. Which one of the following statement is correct about this system?
- A. The system has high coupling between subsystems.**
B. The system has low coupling between subsystems.
C. The system has high cohesion classes in the subsystem.
D. The system has low cohesion between subclasses.
57. Which one the following is incorrect about relationship building block of Unified Modelling Language (UML).
- A.** Dependency is a relationship between two things in which change in one element also affects the other one.
B. Association relationship exists in case of interfaces.

- C. Generalization describes inheritance relationship in the world of objects.
 - D. Realization can be defined as a relationship one element describes some responsibility which is not implemented and the other one implements them.
58. All are true about use case diagram. Except
- A. Describe the functional behavior of the system as seen by the user.
 - B. Use cases are a set of actions, services, and functions that the system needs to perform.
 - C. An actor may be user, external system or physical environment whose models an external entity which communicates with the system.
 - D. An actor has represents multiple users then the actor name must be plural noun.

Woldia University is using Student Information Management System (SIMS) now. Assume this system has not class model document.

Answer question number 59 and 60 based on the above information

59. Obtaining class model from the existing SIMS source code is called _____
- A. Forward Engineering
 - B. Re-engineering
 - C. Reverse Engineering
 - D. Modification Engineering.

60. Which one of the following statement is incorrect about the advantage of obtaining class model from the existing SIMS source code?
- A. To understand the architecture of an existing system.
 - B. It is needed during maintenance when the system fails.
 - C. It is using during upgrade or which is needed to be shift to other operating system.
 - D. It is not applicable for object oriented software development paradigm. But it is applicable for traditional software development strategy.**
61. Which one of the following statement is incorrect about object oriented software development strategy?
- A. Object orientation approach support abstraction at data level**
 - B. Object-oriented approach uses the same language to talk about analysis, design, programming and database design.
 - C. Object orientation approach support abstraction at object level
 - D. Promotion of reusability
62. Which of the following statement is false about subsystem?
- A. Subsystem decompositions help to reduce the complexity of the system.
 - B. In UML subsystems are modeled as deployment diagram.**
 - C. A large system is usually decomposed into subsystems using both, layers and partitions.
 - D. The objects and classes from the object model are the “seeds” for the subsystems.

VI. Computer Programming

63. Quality programming that comes from good program design is necessary for the economic and correct solution of problems. Assume you are the program designer and the program that you design cannot adapt one type of computer to another with minimum modification. So, what need of a good program design is failed?
- A. Reliability
 - B. Efficiency
 - C. Maintainability
 - D. Portability**
64. Which one of the following is **INCORRECT** about an algorithm?
- A. Set of instructions showing the steps of solving the problem.
 - B. All the steps can be dependent on any programming language or machine.**
 - C. The steps must be efficient in time, memory and other resources
 - D. It must produce expected output and can be simple and correct
65. Which one of the following should include under every C++ program?
- A. int
 - B. double
 - C. main ()**
 - D. length ()
66. A symbolic name for memory location in which data can be stored and subsequently recalled is called_____.
- A. Keyword
 - B. Literals
 - C. Variable**
 - D. Reserved words

67. Which one of the following statements provides a way of choosing between a set of alternatives based on the value of an expression?
- A. if statement
 - B. if...else statement
 - C. **switch statement**
 - D. do.... while statement
68. Among the following which one is used to execute block of code known times?
- A. **For loop**
 - B. While loop
 - C. Do...while loop
 - D. Switch
69. The if...else statement can be replaced by which operator?
- A. Bitwise operator
 - B. **Conditional operator**
 - C. Multiplicative operator
 - D. Addition operator
70. In situations where we need to execute body of the loop before testing the condition, we should use_____.
- A. for loop
 - B. while loop
 - C. **do-while loop**
 - D. nested for loop
71. Which one of the following is **INCORRECT** about pointers in C++.
- A. It is a variable that stores the address of another variable.
 - B. It points to the address of another variable.
 - C. It should have the same data type as that of the variable it points to.
 - D. **The reference operator (&) is used to access the value at an address.**
72. Which one of the following statements is **CORRECT** regarding with function?
- A. **A function definition with a missing function prototype.**
 - B. If we define the function above the main method, we must declare it.
 - C. If we define the function below the main method, it is not mandatory to declare it.
 - D. User-defined function definitions are executed without a function call.
73. The header file listed in C++ standard library that contains function prototypes for the stream manipulators that enable formatting of streams of data is?
- A. <iomanip>
 - B. **<fstream>**
 - C. <iostream>
 - D. <exception>

VII. Web Programming

74. Who is making the web standards?
A. Microsoft **B. The World Wide Web Consortium** C. Google D. Mozilla
75. What is the correct HTML for creating a hyperlink?
A. `<a>http://www.google.com</a>`
B. `<a name =" http://www.google.com">Google.com</a>`
C. `<a href=http://www.google.com">Google</a>`
D. `<a url=" http://www.google.com">Google.com</a>`
76. How can you open a link in a new tab/browser window?
A. `<a href="url" target="_blank">`
B. `<a href="url" new>`
C. `<a href="url" target="new">`
D. `<a href="url">new tab</a>`
77. In html, which attribute is used to specify that an input field must be filled out?
A. formvalidate **B. required** C. validate D. placeholder
78. An `<iframe>` is used to display a web page within a web page.
A. There is no such thing as an `<iframe>` B. False C. **True** D. None
79. What is the correct HTML for inserting an image:
A. `<img alt="MyImage">image.gif</img>`
B. `<img href=" image.gif" alt=" MyImage">`
C. `<image src=" image.gif" alt=" MyImage">`
D. ``
80. _____ is a metalanguage used to define new markup languages, and help you to create a language crafted specifically for your application or domain.
A. PHP B. HTML **C. XML** D. JavaScript
81. _____ are the software programs people use to consume the web?
A. Languages **B. Browsers** C. HTTP D. Web Servers
82. How do you create a cookie in PHP?
A. **setcookie()** B. createcookie C. makecookie() D. addcookie
83. How do you create an array in PHP?
A. `$cars = "Volvo", "BMW", "Toyota";`
B. `$ cars [3] = "Volvo", "BMW", "Toyota";`
C. `$cars = array("Volvo", "BMW", "Toyota");`
D. `$cars = array["Volvo", "BMW", "Toyota"];`
84. What is MySQL?
A. A relational database management system
B. Cross-platform, open-source and free
C. Developed, distributed, and supported by Oracle Corporation
D. All
85. Which statement is false?
A. When using the POST method, variables are displayed in the URL.
B. PHP can generate dynamic page content.
C. The if statement is used to execute some code only if a specified condition is true.

D. PHP scripts are executed on the server.

VIII. Fundamentals of Database System

86. Schema normalization is not for?

- a. Reducing the number of tables in the schema.
- b. Eliminating uncontrolled redundancy of data.
- c. Eliminating anomalies that could otherwise occur with inserts.
- d. Eliminating anomalies that could otherwise occur with deletes.

87. Which one of the following statements is used to create a table?

- a. CREATE TABLE table_name (column_name column_type);
- b. CREATE table_name (column_type column_name);
- c. CREATE table_name (column_name column_type);
- d. CREATE TABLE table_name (column_type column_name);

88. What is the difference between Generalization and Specialization?

- a. Generalization and Specialization is like the Top-Down Approach
- b. Generalization and Specialization is like the Bottom-Up Approach
- c. Generalization is like a Bottom-Up approach and Specialization is like a Top-Down Approach
- d. Generalization is like the Top-Down approach and Specialization is like the Bottom-Up Approach

89. A database language which provides users with high privilege command granting privileges and allowing granted privileges by the owner at any time is called_____?

- a. Transaction Control Language C. Data Manipulation Language
- b. Data Control Language D. Data Definition Language

90. A rule assigned to a column in a table that prevents certain types of inconsistency data value from being inserted or placed in the column?

- A. Keys B. Constraint C. Tuple D. Attribute

91. Which one of the following is not Functions of Structured Query Language?

- a. It can execute and retrieve queries against a database
- b. It can insert and update records in a database
- c. It can create stored procedures and views in a database

- d. All of the above
92. _____ is a set of one or more attributes taken collectively to uniquely identify a record?
- a. Candidate key
 - b. **Super key**
 - c. Primary key
 - d. Foreign key
93. Which of the following refers to the level of data abstraction that describes exactly how the data actually stored?
- a. Logical level
 - b. Conceptual level
 - c. **Physical Level**
 - d. None
94. Which of the following is an example of object based logical model?
- A. **Entity Relationship Model**
 - B. Hierarchical Model
 - C. Relational Model
 - D. Network Model
95. Which of the following refers to the number of attributes in a relation?
- A. Cardinality
 - B. **Degree**
 - C. Row
 - D. Domain
96. Which of the following level is considered as the level closed to the end-users?
- A. **External level**
 - B. Internal Level
 - C. Physical Level
 - D. Conceptual Level
97. Which one of the following commands is used to modify a column inside a table?
- A. Drop
 - B. Update
 - C. **Alter**
 - D. Insert

Part 9. Advanced Database System

98. Which of the following is odd?
- A. Locking
 - B. Time stamping
 - C. **Optimistic**
 - D. None
99. A system that manages transactions and controls their access to a DBMS is called _____
- A. **Transaction monitor**
 - B. Transaction processing system
 - C. DBMS
 - D. None
100. Which level of security measures concerned with authorization of database users for access the content at different levels and privileges?
- A. Application level
 - B. **Human level**
 - C. Physical level
 - D. Data base system
101. What is a mechanism for enforcing limits on access to a resource in an environment where there are many threads of execution?

- A. Time stamping
 - B. Concurrency control
 - C. Transaction
 - D. Locking**
102. Which of the following is true about transaction?
- A. Transaction could not be the whole program that can update the entire data base
 - B. Transaction could be a single command that can update the entire data base**
 - C. Transaction do not include changes made in real time to a database
 - D. Transaction is a mechanism for applying the desired modifications to a database
103. Which of the following statement is true about lock?
- A. Lock(X) means data item X is not available to other transactions**
 - B. Lock(X) means data item X is available to other transactions
 - C. Unlock(X) means data item X is not available to other transactions
 - D. A read operation can change the value of a data item
104. Which of the following is the correct match?
- A. Success-Roll back
 - B. Success-Commit**
 - C. Failure- Commit
 - D. Success-Abort
105. Which of the following is the correct statement?
- A. Durability is changes to data are performed as if they are a single operation
 - B. Atomicity refers the database must begin in a consistent state and end in a consistent state
 - C. The integrity of the database cannot be guaranteed without the ACID property**
 - D. Consistency refers transaction changes that are saved to the database cannot be lost even if the system crashes
106. Which of the following is not true?
- A. Begin transaction----abort-----failed----terminated
 - B. Begin transaction----partially committed-----committed----terminated
 - C. Begin transaction----partially committed-----failed----terminated
 - D. Begin transaction----abort-----failed----committed----terminated**
107. Which of the following statement is not true about data base recovery?
- A. Database recovery is the process of restoring database to a correct state in the event of a failure

- B. A database recovery is the process of eliminating the effects of a failure from the database
- C. A database recovery is the process of restoring the last consistent state of the data items.
- D. A database recovery is the process of encouraging the effects of a failure from the database
108. Which of the following recovery facility contains information about all updates to database?
- A. Back up
- B. Check point
- C. Log files
- D. Recovery manager
109. Suppose the given data items $X = 100$, $Y = 50$, and $N = 10$ and T_1 and T_2 are transactions, what is the value of data items of X and Y on T_2 respectively if T_1 followed by T_2 ?
- | | |
|----------------------------|----------------------------|
| T1: Read($X=100$) | T2: T1: Read(X) |
| $X=X+N$ | $X=X+N$ |
| Write($X=X+N$) | Write($X+N$) |
| Read($Y=50$) | Read(Y) |
| $Y=Y+N$ | $Y=Y-N$ |
| Write ($Y=Y+N$) | Write ($Y=Y-N$) |
- A. 110 & 40
- B. 110 & 60
- C. 120 & 50
- D. 120 & 40
110. Among data allocation strategies, which one is copies of one or more partitions are stored at several sites?
- A. Centralized
- B. Partitioned
- C. Replicated
- D. None

Part 10. Computer Security

111. An information system that processes sensitive information is configured to require a valid userid and strong password from any user. This process of accepting and validating this information is known as:
- A. Authentication
- B. Strong authentication
- C. Two-factor authentication

D. Single sign-on

112. An organization wants to prevent SQL and script injection attacks on its Internet web application. The organization should implement a/an:

A. Intrusion detection system

B. Firewall

C. Application firewall

D. SSL certificate

113. CIA is known as:

A. Confidentiality, Integrity, and Availability

B. Computers, Information, and Assets

C. Confidence In Applications

D. Controls, Integrity, and Availability

114. A database administrator (DBA) is responsible for carrying out security policy, which includes controlling which users have access to which data. The DBA has been asked to make just certain fields in some database tables visible to some new users. What is the best course of action for the DBA to take?

A. Implement column-based access controls

B. Export the table to a data warehouse, including only the fields that the users are permitted to see

C. Clone the table, including only the fields that the users are permitted to see

D. Create a view that contains only the fields that the users are permitted to see

115. Public key cryptography is so-named because:

A. It is the world standard for HTTPS

B. It works on all popular computer operating systems

C. It uses an encryption key that can be released to the public

D. The encryption algorithms reside in the public domain

116. A particular encryption algorithm transforms plaintext to ciphertext by XORing the plaintext with the encryption key. This is known as:

A. Electronic codebook

B. Cipher block chaining

C. Block cipher

D. Stream cipher

117. Which of the following statements is correct about the firewall?

- A. It is a device installed at the boundary of a company to prevent unauthorized physical access.
- B. It is a device installed at the boundary of an incorporate to protect it against the unauthorized access.
- C. It is a kind of wall built to prevent files form damaging the corporate.
- D. None of the above.

118. The Data Encryption Standard:

- A. Is used by Secure Sockets Layer (SSL) encryption
- B. Has been replaced by the International Data Encryption Algorithm (IDEA)
- C. Uses a 64-bit encryption key
- D. Uses a 56-bit encryption key

119. A security manager is setting up resource permissions in an application. The security manager has discovered that he can establish objects that contain access permissions, and then assign individual users to those objects. The access control model that most closely resembles this is:

- A. Access matrix
- B. Mandatory access control (MAC)
- C. Discretionary access control (DAC)
- D. Role based access control (RBAC)

120. Which of the following usually observe each activity on the internet of the victim, gather all information in the background, and send it to someone else?

- A. Malware
- B. Spyware
- C. Adware
- D. All of the above